

MATHEMATICS (SINDH BOARD)

CLASS **11th**

CHAPTER **6**

EXERCISE **6.2**

Q1: Evaluate the following :

i. ${}^{20}P_3$

Solution:

$${}^{20}P_3$$

We know that :

$${}^nP_r = \frac{n!}{(n-r)!}$$

$${}^{20}P_3 = \frac{20!}{(20-3)!} = \frac{20!}{17!} = \frac{20 \times 19 \times 18 \times \cancel{17!}}{\cancel{17!}}$$

$$= \frac{20 \times 19 \times 18 \times \cancel{17!}}{\cancel{17!}}$$

$${}^{20}P_3 = 6840 \quad \underline{\text{Ans}}$$

$${}^nP_r = \frac{n!}{(n-r)!}$$

ii. ${}^{16}P_4$

Solution:

$${}^{16}P_4$$

We know that :

$${}^nP_r = \frac{n!}{(n-r)!}$$

$${}^{16}P_4 = \frac{16!}{(16-4)!} = \frac{16!}{12!} = \frac{16 \times 15 \times 14 \times 13 \times \cancel{12!}}{\cancel{12!}}$$

$${}^{16}P_4 = 43680 \quad \underline{\text{Ans}}$$

Q#2: Find the value of n in each of the following:

i. ${}^nP_2 = 30$

Solution:

${}^nP_2 = 30$

We know that:

$${}^nP_r = \frac{n!}{(n-r)!}$$

$$30 = \frac{n!}{(n-2)!}$$

$$30 = \frac{n(n-1)(n-2)!}{(n-2)!}$$

$$30 = n(n-1)$$

$$30 = n^2 - n$$

$$n^2 - n - 30 = 0$$

$$n^2 - 6n + 5n - 30 = 0$$

$$n(n-6) + 5(n-6) = 0$$

$$(n-6)(n+5) = 0$$

$$n-6 = 0$$

$$n = 6$$

$$n+5 = 0$$

$$n = -5 \text{ (not accepted)}$$

$n=6$ is the answer.

ii. ${}^nP_3 = 60$

Solution:

${}^nP_3 = 60$

We know that:

$${}^nP_r = \frac{n!}{(n-r)!}$$

$$60 = \frac{n!}{(n-3)!}$$

$$60 = \frac{n(n-1)(n-2)(n-3)!}{(n-3)!}$$

$$60 = (n^2 - n)(n-2)$$

$$n^3 - 2n^2 - n^2 + 2n - 60 = 0$$

$$n^3 - 3n^2 + 2n - 60 = 0$$

$$n^3 - 5n^2 + 2n^2 - 10n + 12n - 60 = 0$$

$$n(n-5) + 2n(n-5) + 12(n-5) = 0$$

$$(n-5)(n+2n+12) = 0$$

$${}^nP_r = 30$$

$${}^nP_r = \frac{n!}{(n-r)!}$$

$$n! = n(n-1)(n-2)!$$

$$n^2x - 30 = -30n^2$$



$$(n-5)(n+2n+12) = 0$$

$$n-5 = 0$$

$$\boxed{n = 5}$$

$$(n+2n+12) = 0$$

Its factors are not possible then answer will be a complex. then it is not accepted.

vii. ${}^8P_n = 8 \cdot 7 \cdot 6$

Solution:

$${}^8P_n = 336$$

we know that:

$${}^nP_r = \frac{n!}{(n-r)!}$$

$$\therefore r = n$$

$$336 = \frac{8!}{(8-n)!}$$

$$(8-n)! = \frac{8!}{336}$$

$$(8-n)! = \frac{40320}{336}$$

$$(8-n)! = 120$$

$$\therefore 120 = 5!$$

$$(8-n)! = 5!$$

$$(8-n) = 5$$

$$8-5 = n$$

$$\boxed{n = 3} \text{ Ans}$$

iv. ${}^{11}P_n = 11 \cdot 10 \cdot 9$

Solution:

${}^{11}P_n = 990$

we know that:

$${}^nP_r = \frac{n!}{(n-r)!}$$

$$990 = \frac{11!}{(11-n)!} \quad \therefore n=r$$

$$990 = \frac{11!}{(11-n)!}$$

$$(11-n)! = \frac{11!}{990}$$

$$(11-n)! = 40320$$

$$(11-n)! = 8!$$

$$\frac{11-8=n}{\boxed{n=3}}$$

$$\therefore 40320 = 8!$$

v. ${}^{n+2}P_4 : {}^{n+1}P_3 = 5 : 1$

Solution:

$${}^{n+2}P_4 : {}^{n+1}P_3 = 5 : 1$$

$$= \frac{{}^{n+2}P_4}{{}^{n+1}P_3} = \frac{5}{1} \quad \therefore {}^nP_r = \frac{n!}{(n-r)!}$$

$$= \frac{(n+2)!}{(n+2-4)!} \div \frac{(n+1)!}{(n+1-3)!} = 5$$

$$= \frac{(n+2)!}{(n-2)!} \div \frac{(n+1)!}{(n-2)!} = 5$$

$$= \frac{(n+2)!}{(n-2)!} \times \frac{(n-2)!}{(n+1)!} = 5$$

$$= \frac{(n+2)!}{(n+1)!} = 5$$

$$= \frac{(n+2)(n+1)n!}{(n+1)(n!)} = 5$$

$$= n+2 = 5$$

$$= n = 5-2$$

$$\begin{aligned} (n+2)! &= (n+2)(n+1)n! \\ (n+1)! &= (n+1)n! \end{aligned}$$

$$\boxed{n=3} \text{ Ans}$$

$$\text{vi- } {}^n P_4 : {}^{n-1} P_3 = 9 : 1$$

Solution:

$${}^n P_4 : {}^{n-1} P_3 = 9 : 1$$

$$= \frac{{}^n P_4}{{}^{n-1} P_3} = \frac{9}{1}$$

$$\therefore {}^n P_r = \frac{n!}{(n-r)!}$$

$$= \frac{n!}{(n-4)!} \div \frac{(n-1)!}{(n-1-3)!} = 9$$

$$= \frac{n!}{(n-4)!} \div \frac{(n-1)!}{(n-4)!} = 9$$

$$= \frac{n!}{(n-4)!} \times \frac{(n-4)!}{(n-1)!} = 9$$

$$= \frac{n!}{(n-1)!} = 9$$

$$= \frac{n(n-1)!}{(n-1)!} = 9$$

$$= \boxed{n=9} \quad \underline{\underline{\text{Ans}}}$$

$$\therefore n! = n(n-1)!$$

Q#3: In how many ways different batting orders are possible for a cricket team consisting of 11 players.

Solution:

Total Players = 11

Total ways = 11

Total ways = nP_n

$${}^nP_r = \frac{n!}{(n-r)!} = {}^nP_n = \frac{11!}{(11-11)!} = \frac{11!}{0!} \because 0! = 1$$

$${}^nP_n = 11!$$

$$= 39916800 \text{ Ans}$$

Thus 39916800 ways different batting orders are possible.

Q#4: In how many ways can two English books three chemistry books and four Physics books be arranged on a shelf so that all the books of same subject are together?

Solution:

English books = 2

Chemistry books = 3

Physics books = 4

Total books types = 3

Books on a shelf are arranged = $2! 3! 4! 3!$

$$= 2 \times 3 \times 2 \times 1 \times 4 \times 3 \times 2 \times 3 \times 2$$

$$\text{Total ways} = 1728$$

Q#5: How many different arrangement can be formed of the word 'MATRIX' using all letters? Also find the numbers of arrangement if each of them start with 'R'.

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Solution:

In word MATRIX = 6 numbers are used.

Different arrangement can be formed of the word MATRIX is 6!

$$\text{Total arrangement} = 6! \\ = 720$$

If R is the first letter then number of arrangement form remaining five letters = 5!

5 letters can be arranged in 5! ways.

120 total ways when R is the first letter.

Q#6: How many signals can be given by 8 flags of different colours when 5 of them are used at a time?

Solution:

Total flags = 8

Used at a time = 5

Total signals can be = 8P_5

We know that:

$${}_nP_r = \frac{n!}{(n-r)!}$$

$${}^8P_5 = \frac{8!}{(8-5)!} = \frac{40320}{3!}$$

$${}^8P_5 = 6720$$

Thus, 6720 signal can be given by 8 flags of different colours when 5 of them are used at a time. Ans

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Q#7: A fair coin is tossed three times. How many outcomes are possible? Find by fundamental principle and illustrate using tree diagram.

Solution:

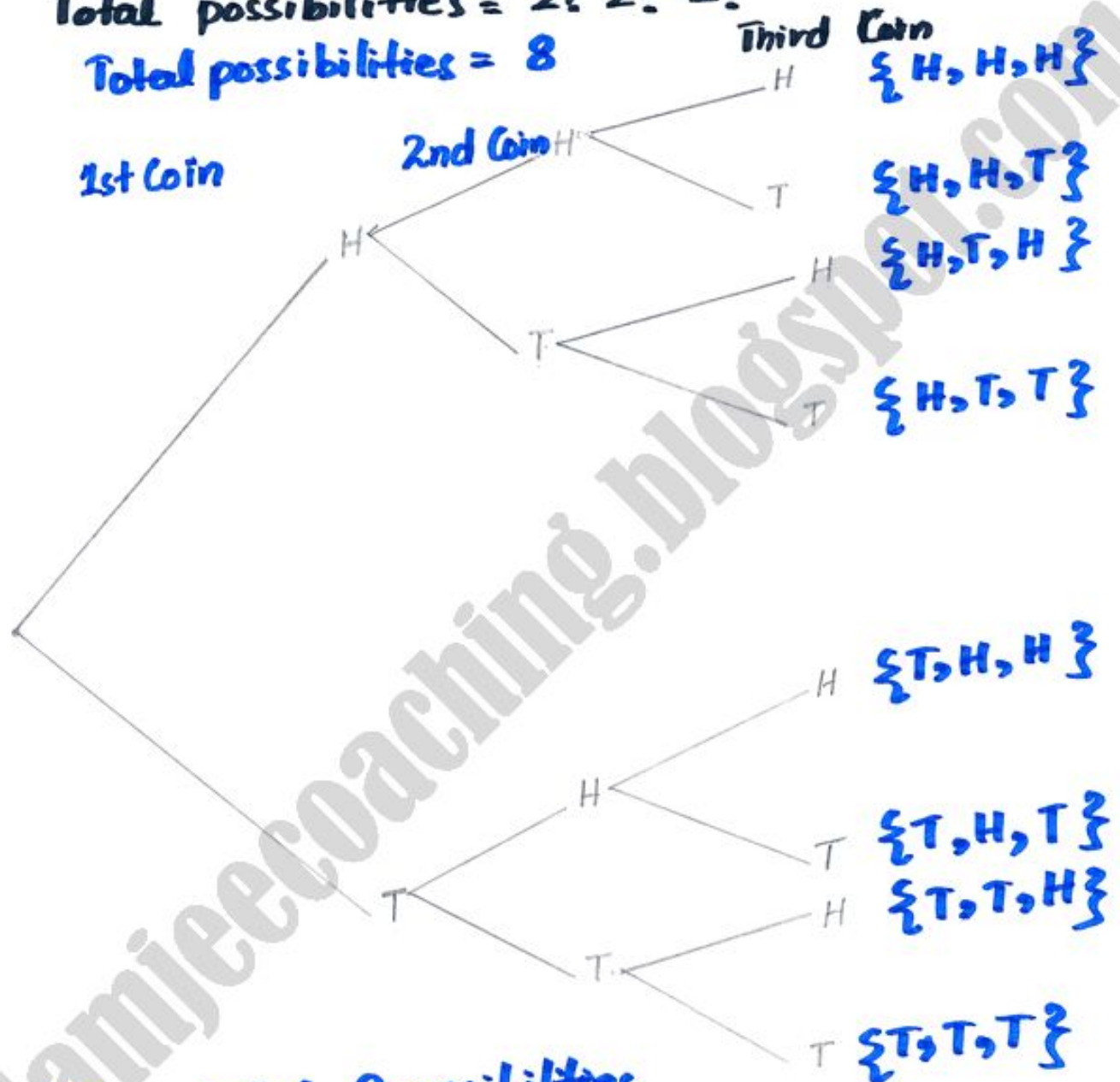
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Coin is tossed = 3 times.

Total possibilities = 2 (at a time)

Total possibilities = $2! 2! 2!$

Total possibilities = 8



Thus, total 8 possibilities

are $\{H, H, H, HHT, HTH, HTT, THH, THT, TTH, TTT\}$

Q# 8: How many (five) 5-digit numbers can be formed, without repeating any digit:

i. 5, 6, 7, 8, 9

ii. 1, 2, 3, 5, 7, 9

iii. 2, 4, 6, 8, 0

Solution:

i. 5, 6, 7, 8, 9

Solution:

First place can be filled by = 5 ways

Second place can be filled by 4 ways

Third place can be filled by 3 ways

Fourth place can be filled by 2 ways

Fifth place can be filled by 1 way

Total 5 (digit) number can be formed

by $5 \times 4 \times 3 \times 2 \times 1$ ways.

Total 5-digit number can be formed =

120 numbers are formed.

$$\therefore 0! = 1$$

$${}^5P_5 = \frac{n!}{(n-r)!} = \frac{5!}{(5-5)!} = 120$$

ii. 1, 2, 3, 5, 7, 9

Solution:

Here Total numbers = 6

Used at a time = 5

$${}^6P_5 = \frac{6!}{(6-5)!} = 720$$

iii. 2, 4, 6, 8, 0

$$\text{Total number} = {}^5P_5 - {}^4P_4 = 96 \text{ Ans}$$

Solution:

Total number = 5

but 0 is not used in first place then

first place can be filled by 4 ways.

Second place can be filled by 4 ways.

Third place can be filled by 3 ways.

Fourth place can be filled by 2 ways.

Fifth place can be filled by 1 way.

$$\text{Total numbers} = 4 \times 4 \times 3 \times 2 \times 1 = 96$$

Q#9: Prove that:

$$i. {}^n P_r = {}^{n-1} P_r + r \cdot {}^{n-1} P_{r-1}$$

$$ii. {}^n P_r = n \cdot {}^{n-1} P_{r-1}$$

Proof:

$${}^n P_r = {}^{n-1} P_r + r \cdot {}^{n-1} P_{r-1}$$

Taking R.H.S:

$$= {}^{n-1} P_r + r \cdot {}^{n-1} P_{r-1}$$

$$\therefore {}^n P_r = \frac{n!}{(n-r)!}$$

$$= \frac{(n-1)!}{(n-1-r)!} + \frac{r \cdot (n-1)!}{(n-1-(r-1))!}$$

$$= \frac{(n-1)!}{(n-1-r)!} + \frac{r \cdot (n-1)!}{(n-1-r+1)!}$$

$$= \frac{(n-1)!}{(n-1-r)!} + \frac{r \cdot (n-1)!}{(n-r)!}$$

$$= \frac{(n-1)!}{(n-1-r)!} + \frac{r \cdot (n-1)!}{(n-r)(n-r-1)!}$$

$$= \frac{(n-1)!}{(n-1-r)!} \left\{ 1 + \frac{r}{(n-r)} \right\}$$

$$= \frac{(n-1)!}{(n-1-r)!} \left\{ \frac{n-r+r}{(n-r)} \right\}$$

$$= \frac{(n-1)!}{(n-1-r)!} \times \frac{n}{(n-r)}$$

$$= \frac{n(n-1)!}{(n-r)(n-1-r)!}$$

$$= \frac{n!}{(n-r)!}$$

$$= {}^n P_r$$

$${}^n P_r = {}^{n-1} P_r + r \cdot {}^{n-1} P_{r-1}$$

L.H.S = R.H.S

$$\therefore n(n-1)! = n!$$

$$\therefore (n-r)(n-1-r) = (n-r)!$$

$$\therefore \frac{n!}{(n-r)!} = {}^n P_r$$

Hence Proved.

$$\text{ii- } {}^n P_r = n \cdot {}^{n-1} P_{r-1}$$

Proof:

$${}^n P_r = n \cdot {}^{n-1} P_{r-1}$$

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Taking R.H.S

$$= n \cdot {}^{n-1} P_{r-1}$$

$$= n \cdot \frac{(n-1)!}{(n-1-(r-1))!}$$

$$= \frac{n \cdot (n-1)!}{(n-1-r+1)!}$$

$$= \frac{n(n-1)!}{(n-r)!}$$

$$= \frac{n!}{(n-r)!}$$

$$= {}^n P_r$$

$$\text{L.H.S} = \text{R.H.S}$$

$${}^n P_r = n \cdot {}^{n-1} P_{r-1}$$

$$\therefore {}^n P_r = \frac{n!}{(n-r)!}$$

$$= n(n-1)! = n!$$

$$\therefore \frac{n!}{(n-r)!} = {}^n P_r$$

Hence Proved

Q#10: How many arrangements can be formed using all the letters of the following words:

i. STATISTICS

ii. PLANE

iii. OBJECT

iv. FASTENS

v. MATHEMATICS

vi. ASSISTANCE

i. STATISTICS.

Solution:

Total letters = 10

Repeated letters are S = 3, T = 3, I = 2

$$\text{Total arrangement} = \frac{10!}{3! 3! 2!}$$

$$= \frac{3628800}{6 \times 6 \times 2}$$

$$= \frac{362880}{12}$$

$$\text{Total arrangements} = 30240$$

Thus, Total arrangements can be formed using letter STATISTICS is 30240.

ii. PLANE

Solution:

Total letters = 5

Repeated letters are not present.

$$\text{Total arrangements} = 5!$$

$$\text{Total arrangements} = 120$$

Thus, the total arrangement can be formed using letter PLANE is 120.

iii- OBJECT

Solution:

Total letters = 6

Repeated letters are not present.

Total arrangements = $6!$

Total arrangements = 720

Thus, the total arrangements can be formed using letter OBJECT is 720.

iv- FASTENS

Solution:

Total letters = 7

Repeated letter is S = 2

$$\begin{aligned}\text{Total arrangements} &= \frac{7!}{2!} \\ &= \frac{5040}{2}\end{aligned}$$

$$= 2520$$

Thus, the total arrangements can be formed using letter FASTENS is 2520.

v- MATHEMATICS

Solution:

Total letters = 11

Repeated letters are M = 2, A = 2, T = 2

$$\begin{aligned}\text{Total arrangements} &= \frac{11!}{2!2!2!} \\ &= \frac{39916800}{2 \times 2 \times 2} \\ &= \frac{39916800}{8} \\ &= 4989600\end{aligned}$$

Thus, the total arrangement can be formed using letter MATHEMATICS is 4,989,600

Q#11: How many 8-digits numbers can be formed from the digits 2, 2, 2, 3, 3, 3, 5, 6? How many among these are greater than 60,000,000?

Solution:

8 digit numbers formed from 2, 2, 2, 3, 3, 3, 5, 6 are:

$$\text{Total numbers} = 8$$

Repeated numbers are 2 = 3 times, 3 = 3 times.

$$\text{then total number} = \frac{8!}{3! 3!}$$

$$\text{Total 8 digit numbers} = \frac{40320}{6 \times 6}$$

$$\text{Total 8 digit numbers} = 1120$$

Total 8 digit numbers formed when 6 is placed at first place = $\frac{7!}{3! 3!}$

$$= \frac{5040}{6 \times 6} = 140$$

Total 8 digit numbers formed when 6 is at first place is 140.

Q#12: How many distinct permutations of letters of the word "ESSENTIAL" are possible? How many will have the two S's i. together ii. separate.

Solution:

$$\text{Total letters} = 9$$

$$\text{Repeated letters} = S = 2, E = 2$$

i. When two S's are together:

$$\text{Total letters} = 8$$

$$\text{Total Permutation} = \frac{8!}{2!} = \frac{40320}{2} = 20160$$

ii. When two S's are separate:

$$\text{Total letters} = 9$$

$$\text{Total Permutation} = \frac{9!}{2! 2!} - \frac{8!}{2!}$$

$$\text{Total Permutation} = 90720 - 20160$$

$$= 70560 \quad \underline{\underline{\text{Ans}}}$$

Q#16: How many numbers greater than 23,000 can be formed from the digits 1, 2, 3, 5, 6 if digits may repeat?

Solution:

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The first place is filled by

4 ways using 2, 3, 5, 6.

and the second place is filled by 3 ways.

the third place is filled by 5 ways.

the fourth place is filled by 5 ways

the fifth place is filled by 5 ways.

$$(4)(3)(5)(5)(5)$$

$$1500.$$

2	3			
4	3	5	5	5

Q#17: Find the numbers of 5-digits numbers that can be formed from the digits 1, 2, 4, 6, 8 (when no digit is repeated), but

i. the digits 2 and 8 are next to each other.

ii. the digits 2 and 8 are not next to each other.

Solution:

(i) Arrangement of the digits 2 and 8 are next to each other then:

$$28, 82 = 2!$$

$$\text{No. of arrangements} = 4! \cdot 2!$$

$$\text{No. of arrangements} = 48$$

(ii) The digits 2 and 8 are not next to each other then

$$\text{Total arrangements} = {}^n P_r = \frac{n!}{(n-r)!}$$

$$n=5, r=5$$

$${}^5 P_5 = \frac{5!}{(5-5)!} = 120$$

$$\text{No. of arrangement} = \text{Total arrangements} - 48$$

$$= 120 - 48$$

The digits 2 and 8 are not next to each other the no. of arrangements are 72. Ans

Q#18: How many 6-digits numbers can be formed without repeating any digit from the digits 0, 1, 2, 3, 4, 5? In how many of them will 0 be at the tens place?

Solution:

Total digit = 6 digits.

0 is not placed on first place:

$$n=5, r=5$$

$$\text{Total dig numbers can be formed} = {}^n P_r =$$

$$n=6, r=6$$

$$= {}^6 P_6 = {}^5 P_5$$

$$= 720 - 120$$

$$= 600$$

$$\text{Total digits} = 5, r=5$$

$${}^n P_r = \frac{n!}{(n-r)!} = {}^5 P_5 = \frac{5!}{(5-5)!} = 120$$

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Q#19: How many 5-digits multiples of 5 can be formed from the digits 2,3,5,7,9, when digits may repeat?

Solution:

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Total numbers are = 5

Digit may repeat but

here multiples of 5 can be formed then the unit place is fixed with 5 digit using 5 then, we used at ~~unit~~ tenth place, hundred place, thousandth place and Tenth thousandth place with 5 digits. using 2,3,5,7,9 (because digits may repeat)

then,

$$= 5 \times 5 \times 5 \times 5$$

$$= 625$$

625 ~~a~~ multiples of 5 can be formed.

Ans

Q#13: In how many ways, can 8 persons be seated around a round table?

Solution:

Total arrangements in a circle = $(n-1)!$

Here $n=8$

$$\begin{aligned}\text{Total arrangements} &= (8-1)! \\ &= 7! \\ &= 5040\end{aligned}$$

Thus, 5040 ways are 8 can 8 persons be seated around a round table.

Q#14: In how many ways, 7 monkeys be arranged in a circular key ring?

Solution:

Total arrangements in a key ring =

$$= \frac{(n-1)!}{2}$$

Here $n=7$

$$\begin{aligned}\text{Total arrangements in a key ring} &= \frac{(7-1)!}{2} \\ &= \frac{6!}{2} = 360\end{aligned}$$

360 ways can we arranged 7 keys in a circular key ring.

Q#15: How many 3-digit numbers can be formed by using each one of the digits 2, 3, 5, 7, 9 only once?

Solution:

Total digits = 5 = n , $r=3$

Digits may not be repeated:-

Digits can be formed = ${}^n P_r$

$${}^5 P_3 = \frac{(5)!}{(5-3)!} = \frac{120}{2}$$

$${}^5 P_3 = 60$$

$${}^n P_r = \frac{(n)!}{(n-r)!}$$